DISCUSSION

IDENTIFICATION OF PMM2 AND THE DISEASE-ASSOCIATED VARIANT

The substitution of nucleotides C to T in the position 710 (c.710C>T) caused a missense variant in which the gene codifies for the Methionine amino acid instead of Threonine. Although they are both uncharged, they have different properties. Threonine is polar and has an hydroxyl group (OH-) which allows it to form hydrogens bonds with Thr16. But, it is replaced by Methionine which doesn't have an hydroxyl group required for the interaction disrupting protein interactions. A study made in 2022 by Briso-Montiano et al. determined that changing a Threonine by a Methionine indeed has an effect on the stability of the protein, its interactions, and the enzymatic activity leading to malfunction through the loss of the hydrogen bond. However, the variant kept a similar structure to the wild-type protein , so a change in the sequence doesn't necessarily alter the structure of the protein.
We identified that 237 is located inside the alpha-helix, where a Mg2+ binding site is located. Mg2+ helps to conduct the enzymatic reaction of PMM2. The substitution could affect the molecular interactions and structure. However, we don't have enough proof to demonstrate that the mutant affects the Mg 2+ union. Furthermore, the results of RMSD say that the structure has been kept quite alike.
PMM2 produces mannose-1-phosphate from mannose-6-phosphate important for GDP-mannose to take part in the glycosylation. An altered PMM2 enzymatic activity can affect the glycosylation contributing to the congenital disorder (PMM2-CDG).

With all the analysis, the substitution can affect the stability of the protein and its activity. The consequences of the variant T237M are involved in a malfunction of the protein and glycosylation provoking an insufficiency for the organism leading to the disease PMM2-CDG.

STRUCTURAL COMPARISON OF WILD-TYPE AND T237M PMM2

From the superposition of 7O4G and 7O58 we obtained a low RMSD (0.352 Å). This means that the positions of the atoms in the main structure of the protein are very similar with little differences. The backbone is the main chain of the protein, and consequently the difference between them it's also low so the structure has little variations having a similar protein structure. The structure of the PMM2 and T237M are similar, so we can deduce that the change of aminoacid didn't have a noticeable impact on the protein. However, it doesn't mean that there hasn't been any modifications between the interactions of the chains. Or we shouldn't think that such subtle changes didn't provoke consequences related to the functionality and enzymatic activity.

CROSS-SPECIES COMPARISON OF GENE AND PROTEIN SEQUENCES

The comparisons with different species, Leishmania mexicana and Vicugna pacos gave us an idea of the conservation of the sequences allowing us to measure its importance. Starting with V.pacos, we got a 91% identical sequence of nucleotides with the homo sapiens. But only 62 % of the sequence could have been aligned. In this case, we know that more than

the majority part of the sequence is very similar to homo sapiens. For Leishmania mexicana, it was possible only to align the 21 % of the sequence, and from this sequence a 70 % is similar to the Homo sapiens. Regarding the protein, 87 % of the sequence has been able to align with the human. A 90 % have the same positions of the amino acids in both sequences, and which includes not only identical but different with similar chemical properties has a 92%. So, it makes sense that protein positives have a higher percentage than the identity. On the other hand, we aligned 97 % of the sequence of Homo sapiens with Leishmania mexicana. And we obtained that 57 % of the sequence of amino acids are similar and 71% of the sequences with different positions but same chemical properties. .

Overall,we obtained a bigger similitude of the human nucleotide sequences with the Vicugna Pacos. Because we had a bigger sequence to study and almost all of them were identical.

Another important aspect is that the three species in the position 11-17 of PMM2 had different nucleotides but in result they encoded for the same amino acid. This means that in this position that specific amino acid has an importance for the protein, structurally or functionally, which leads to that the three different species through evolution conserved the same sequence of amino acids (FDVDGTL). The fact that the sequences are in three different species means that it's important to have them in this order to maintain the function of the protein.

In the motif, there are Asp12 and Asp14, aminoacids that participate in the magnesium binding. They were conserved in the three species so it suggests that these residues are functionally important. But again, conservation gives us evidence that the regions are significant but we would need more direct experimentation to give a certain conclusion.

STRUCTURAL COMPARISON BETWEEN HUMAN AND L.MEXICANA

The RMSD of 1,66 Å tells us that the structure of the PMM2 L.Mexicana protein and the human is similar, despite having only 56,4 % of amino acid sequence identity. It suggests that having a quite large percent of different amino acids doesn't mean that the structure of the protein would result in drastic changes. In our case, although having differences in the sequence, the structure of PMM2 L.Mexicana and human PMM2 have a similar fold.

The three-dimensional structure is important to deduce how the molecules interact with each other, but having similar structures doesn't imply that the activity of the protein or activity is going to be the same. Local interactions and differences in the side chains of amino acids might affect the functionality of the protein.

The residue 237 is located inside the alpha-helix in both structures with a local backbone displacement of 1.465 Å, indicating a bit of difference of position between the two structures. The alpha-helix is a secondary form of the protein shaped in a spiral way, contributing to protein folding stabilised by interactions. However, having a variation in the position in this section doesn't demonstrate an alteration in its stability.

Regarding the comparison with the Vicugna pacos, which we saw earlier that it's very similar to the human PMM2 sequence, the protein sequence which included our residue 237 wasn't available. So, we couldn't study if the residue is conserved nor compare it structurally.

LIMITATIONS

We interpreted the results and concluded that the L.Mexicana has a similar structure to the human. However, we couldn't make a proper analysis due to several limitations that we should consider. First of all, the RMSD and local backbones provide only information about the differences in the positions of the protein but not the function. We don't have any analysis that determines the enzymatic activity or function. So we can't determine that the differences in structure affect the PMM2 function.

Also, in the comparison of the species we hadn't available all the regions of the sequences, so we depended on what we obtained. For instance, we only have 21% of the Leishmania Mexicana sequence aligned with the human leading to our conclusions being limited. Furthermore, although the comparisons with the PMM2 structure revealed a similarity between the species we would need additional analysis of the activity to determine the function.

Lastly, our analyses can't fully confirm the consequences in functionality of the variant T237M leading to the disease. Further experimentation of the stability would be needed to clarify how the change affects the function.